

AI Health Report Analyzer

Project Documentation | April 2026

1. Project Overview

AI Health Report Analyzer is a web application that helps users understand medical reports by extracting report text and converting complex lab results into clear health insights. Users can upload PDFs, images, or text files and receive simplified explanations, risk indicators, and lifestyle suggestions.

2. Problem Statement

Many patients receive lab reports filled with technical terms and numbers that are difficult to interpret. This creates confusion, anxiety, and delayed follow-up actions. The project solves this by making reports easier to understand instantly.

3. Solution

The application reads uploaded reports, detects important health values such as thyroid, cholesterol, glucose, blood pressure, and hemoglobin, compares them with standard ranges, and generates easy-to-read recommendations.

4. Tools & Technologies Used

- Python – core programming language
- Streamlit – web interface
- PyPDF2 / pdfplumber – PDF text extraction
- Tesseract OCR / pytesseract – image text recognition
- Pillow / OpenCV – image preprocessing
- Pandas – data handling
- Plotly – charts and health score visualization

5. Core Features

- Upload PDF, image, or text files
- OCR for scanned reports
- Detection of common medical markers
- Plain-language explanations
- Diet and exercise suggestions
- Health score dashboard

6. System Workflow

1. User uploads file
2. File type detected
3. Text extracted from document
4. Values identified using rules/patterns
5. Results compared with reference ranges
6. Insights displayed in dashboard

7. Problems Faced & Solutions

- OCR accuracy issues on blurry images → Added preprocessing and manual text option.
- Some PDFs had no readable text → Added fallback extraction methods.
- False value detection → Improved regex patterns and validation rules.
- Deployment dependency issues → Added packages configuration and lightweight libraries.

8. Outcomes

The final product provides a free, easy-to-use prototype showing how AI and automation can simplify healthcare information for everyday users.

9. Future Improvements

- More medical conditions
- Report history tracking
- Export to PDF summary
- LLM-based explanations
- Multi-language support

10. Conclusion

This project demonstrates how practical AI tools can solve real user problems by improving access to understandable healthcare information through automation, OCR, and intelligent analysis.